

GEOGRAPHY

INDIAN GEOGRAPHY



INDIAN CLIMATE

MODULE - 5

SOIL TYPES

- **BLACK SOILS**
- **RED SOILS**

BLACK SOILS



MAJOR SOIL TYPES OF INDIA

2

BLACK SOILS



Black soil as found in Deccan Plateau

1

The black soils are also called **regur soils** and **black cotton soils** because cotton is the most important crop sown in this soil.

2

Several theories have been put forward regarding the origin of this group of soils but most pedologists believe that these soils have been **formed due to solidification of lava spread over large areas during volcanic activity in the Deccan Plateau thousands of years ago.**

MAJOR SOIL TYPES OF INDIA

2

BLACK SOILS



3 The parent material for most of the black soil are the **volcanic rocks** that were formed in the **Deccan Plateau**.

4 In **Tamil Nadu**, **gneisses** and **schists** form the parent material.

5 These soils are found in the regions of **high temperature** and **low rainfall**. It is, therefore, a soil group typical to the **dry and hot regions** of the **Peninsula**.

MAJOR SOIL TYPES OF INDIA

2

BLACK SOILS

CHARACTERISTIC OF BLACK SOILS



Cracked black soil

1 The black soil is highly retentive of moisture. It swells greatly on accumulating moisture.

2 Strenuous effort is required to work on such soil in rainy season as it gets very sticky.

3 In summer, the moisture evaporates, the soil shrinks and is seamed with broad and deep cracks.

MAJOR SOIL TYPES OF INDIA

2

BLACK SOILS

CHARACTERISTIC OF BLACK SOILS

- 4 This **cracks permit oxygenation** of the soil to sufficient depths.



- 5 The soil is **self-ploughing in nature**. This is because the loosened soil particles from the dried up soils fall into the cracks.

- 6 The soil “swallows” itself and produces a ploughing effect.

- 7 Because of this **character of slow absorption and loss of moisture**, the black soil retains the moisture for a **very long time**, which helps the crops, especially, the rain fed ones, to sustain even during the dry season.

MAJOR SOIL TYPES OF INDIA

2

BLACK SOILS

COLOUR OF BLACK SOILS

1

The black colour has been attributed by some scientists to the presence of a small proportion of titaniferous magnetite or even to iron and black constituents of the parent rock.



Blackness of parent rock



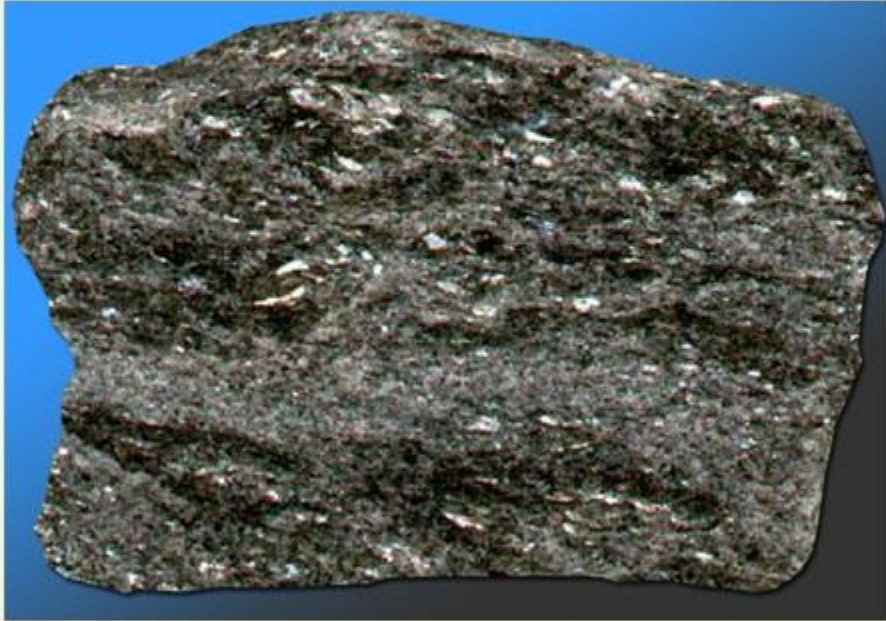
Titaniferous Magnetite

MAJOR SOIL TYPES OF INDIA

2

BLACK SOILS

COLOUR OF BLACK SOILS



Crystalline Schists

2

The black colour of this soil may even be derived from **crystalline schists and basic gneisses** such as in Tamil Nadu and parts of Andhra Pradesh.

3

Various tints of the black colour such as deep black, medium black, shallow black, a mixture of red and black may be found in this group of soils.

MAJOR SOIL TYPES OF INDIA

2

BLACK SOILS

CHEMICAL COMPOSITION



- 1 A typical black soil is highly argillaceous with a large clay factor, 62 per cent or more without gravel or coarse sand.
- 2 Chemically, the black soils are rich in lime, iron, magnesia and alumina. They also contain potash.
- 3 But they lack in phosphorous, nitrogen and organic matter. The color of the soil ranges from deep black to grey.

MAJOR SOIL TYPES OF INDIA

2

BLACK SOILS

DISTRIBUTION OF BLACK SOILS



1

This soil is found in the **regions of high temperature and low rainfall**. It is therefore a soil group of the dry and hot regions of the Peninsula.

2

Geographically black soils are spread over 46 lakh sq km (**16.6 per cent of the total area**) across Maharashtra, Madhya Pradesh, Karnataka, Telangana, Tamil Nadu, Andhra Pradesh, and Gujarat.

MAJOR SOIL TYPES OF INDIA

2

BLACK SOILS

CROPS IN BLACK SOILS



- 1 These soils are **best suited for cotton crop**. Hence these soils are called as black cotton soils.
- 2 Other major crops grown on the black soils include **wheat, jowar, linseed, tobacco, castor, sunflower and millets**.
- 3 Rice and sugarcane are equally important where irrigation facilities are available.
- 4 Large varieties of vegetables and fruits are also successfully grown on the black soils.
- 5 This soil has been used for growing a variety of crops for years without adding fertilizers and manures, with little or no evidence of exhaustion.

RED SOILS



MAJOR SOIL TYPES OF INDIA

3

RED SOILS

1 Most of the red soils have come into existence due to the weathering of ancient crystalline and metamorphic rocks.



Quartzites

2 The main parent rocks are acid granites, gneisses, and quartzites. They occupy about 3.5 lakh sq km (10.6 per cent) of the total area of the country.



Granites

MAJOR SOIL TYPES OF INDIA

3

RED SOILS

CHARACTERISTIC OF RED SOILS



1 The texture of these soils can vary from sand to clay, the majority being loams.

2 On the uplands, the red soils are poor, gravelly, and porous. But in the lower areas, they are rich, deep dark and fertile.

3 These soils mostly occur in the regions of low rainfall.

MAJOR SOIL TYPES OF INDIA

3

RED SOILS

CHEMICAL COMPOSITION



- 1 They are acidic to a certain extent mainly due to the nature of the parent rocks.
- 2 They are poor in lime, magnesia, phosphates, nitrogen and humus.
- 3 They are fairly rich in potash and potassium.

MAJOR SOIL TYPES OF INDIA

3

RED SOILS

COLOUR OF RED SOILS



- 1 The red colour is due to a wide diffusion of iron in crystalline and metamorphic rocks.
- 2 It looks yellow when it occurs in a hydrated form.
- 3 The fine-grained red and yellow soils are normally fertile, whereas coarse-grained soils found in dry upland areas are poor in fertility.
- 4 Ordinarily the surface soils are red while the horizon below gets yellowish colour.

MAJOR SOIL TYPES OF INDIA

3

RED SOILS

DISTRIBUTION OF RED SOILS



1 These soils are spread on almost the whole of Tamil Nadu.

2 Other regions with red soil include parts of Karnataka, south-east of Maharashtra, Telangana, Andhra Pradesh, Madhya Pradesh, Chhattisgarh, Odisha, Chota Nagpur plateau; parts of south Bihar, West Bengal, Uttar Pradesh and parts of North-Eastern states.

MAJOR SOIL TYPES OF INDIA

3

RED SOILS

CROPS OF RED SOILS



1

The red soils are mostly loamy and hence cannot retain water like the black soils.

2

The red soils, with the proper use of fertilizers and irrigation techniques, give good yield of cotton, wheat, rice, pulses, millets, tobacco, oil seeds, potatoes and fruits.

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